(i) Properties, graphs, theorems on inverse trigonometric functions will be studied in higher secondary second year.

(ii) Inverse trigonometric functions are much useful in the evaluation of some integrals which will be studied later.

Example 3.72 Find the principal value of (i) $\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$, (ii) $\text{cosec}^{-1}\left(\frac{2}{\sqrt{3}}\right)$, (iii) $\tan^{-1}\left(\frac{-1}{\sqrt{3}}\right)$.

Solution:

(i) Let $\sin^{-1}\left(\frac{\sqrt{3}}{2}\right) = y$, where $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$

$\Rightarrow \sin y = \frac{\sqrt{3}}{2} = \sin\frac{\pi}{3} \Rightarrow y = \frac{\pi}{3}$

Thus, the principal value of $\sin^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{3}$

(ii) Let $\text{cosec}^{-1}\left(\frac{2}{\sqrt{3}}\right) = y$, where $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$

$\Rightarrow \text{cosec}\, y = \frac{2}{\sqrt{3}} = \sin y = \frac{\sqrt{3}}{2}$

Thus, the principal value of $\text{cosec}^{-1}\left(\frac{2}{\sqrt{3}}\right) = \frac{\pi}{3}$

(iii) Let $\tan^{-1}\left(\frac{-1}{\sqrt{3}}\right) = y$, where $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$

$\tan y = -\frac{1}{\sqrt{3}} \Rightarrow \tan y = \tan\left(-\frac{\pi}{6}\right) \Rightarrow y = -\frac{\pi}{6}$

Thus the principal value of $\tan^{-1}\left(\frac{-1}{\sqrt{3}}\right) = -\frac{\pi}{6}$.

Exercise - 3.11

1. Find the principal value of (i) $\sin^{-1}\frac{1}{\sqrt{2}}$ (ii) $\cos^{-1}\frac{\sqrt{3}}{2}$ (iii) $\text{cosec}^{-1}(-1)$ (iv) $\sec^{-1}\left(-\sqrt{2}\right)$ (v) $\tan^{-1}\left(\sqrt{3}\right)$.
2. A man standing directly opposite to one side of a road of width x meter views a circular shaped traffic green signal of diameter a meter on the other side of the road. The bottom of the green signal is b meter height from the horizontal level of viewer's eye. If α denotes the angle subtended by the diameter of the green signal at the viewer's eye, then prove that

$$\alpha = \tan^{-1}\left(\frac{a+b}{x}\right) - \tan^{-1}\left(\frac{b}{x}\right).$$